

NEPATHYA COLLEGE
(Tribhuvan University)
Tilottama-5, Manigram, Rupandehi



Lab Report of Compiler Design And Construction
Bachelor in Computer Science And Information Technology
(BSc.CSIT)
Tribhuvan University
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Lab 1 output:

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
Enter comment:Hello
It is not a commentxuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
Enter comment:/*Sumanshrestha*/
It is a comment
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
Enter comment://Suman Shrestha
It is a comment
```

Lab 2 Output:

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPPractical$ ./a.out
Enter a string:aaaaabbbb
aaaaabbbb is accepted under rule 'a*b+'
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPPractical$ ./a.out
Enter a string:bbbbaaab
-bbbbbaaab is not recognized
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPPractical$ ./a.out
Enter a string:abb
abb is accepted under rule 'abb'
```

Lab 3 Output:

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
Enter an identifier:area_12no
Valid identifier
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
Enter an identifier:_sum5
Valid identifier
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
Valid identifier
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
Enter an identifier:var@num
Not a valid identifier
```

Lab 4 Output:

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ cat aa.txt
```

```
int main
{
    int age = 20;
    float salary = 50000;
    char grade = a;

    if (age > 18)
    {
        salary = salary + 5000;
    }

    return 0;
}
```

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ gcc Lab-4.c
```

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
```

```
int is keyword
main is identifier
int is keyword
age is identifier
= is operator
20 is identifier
float is keyword
salary is identifier
= is operator
50000 is identifier
char is keyword
grade is identifier
= is operator
a is identifier
if is keyword
age is identifier
18 is identifier
salary is identifier
= is operator
salary is identifier
+ is operator
5000 is identifier
return is keyword
0 is identifier
```

-

Lab 5 Output:

```
xuman-stha@xumanstha-LOQ-15IRH8:~/Documents/CDCPractical$ gcc Lab-5.c
```

```
xuman-stha@xumanstha-LOQ-15IRH8:~/Documents/CDCPractical$ ./a.out
```

```
How many number of productions ? :5
```

```
Enter productions Number 1 : S=L=R
```

```
Enter productions Number 2 : S=R
```

```
Enter productions Number 3 : l=*R
```

```
Enter productions Number 4 : ^C
```

```
xuman-stha@xumanstha-LOQ-15IRH8:~/Documents/CDCPractical$ gcc Lab-5.c
```

```
xuman-stha@xumanstha-LOQ-15IRH8:~/Documents/CDCPractical$ ./a.out
```

```
How many number of productions ? :5
```

```
Enter productions Number 1 : S=L=R
```

```
Enter productions Number 2 : S=R
```

```
Enter productions Number 3 : L=*R
```

```
Enter productions Number 4 : L=a
```

```
Enter productions Number 5 : R=L
```

```
Find the FIRST of :S
```

```
FIRST(S)= { * a }
```

```
press 'y' to continue : y
```

```
Find the FIRST of :a
```

```
FIRST(a)= { a }
```

```
press 'y' to continue : y
```

```
Find the FIRST of :*R
```

```
FIRST(*)= { * }
```

```
press 'y' to continue : xuman-stha@xumanstha-LOQ-15IRH8:~/Documents/CDCPractical$ █
```

Lab 6 Output:

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ gcc Lab-6.c
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
Enter the no. of productions: 5
Enter 5 productions
Production with multiple terms should be give as separate productions
R=aS
R=(R)S
S==RS
S=aRS
S=*S
Find FOLLOW of -->R
FOLLOW(R) = { $ ) + a * }
Do you want to continue(Press 1 to continue....)?1
Find FOLLOW of -->S
FOLLOW(S) = { $ ) + a * }
Do you want to continue(Press 1 to continue....)?^C
^C
```

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ gcc Lab-7.c
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
```

Enter the input string: i*i+i

Stack Input

```
$ b t i*i+i$
$ b c f i*i+i$
$ b c i i*i+i$
$ b c f * *i+i$
$ b c i i+i$
$ b +i$
$ b t + +i$
$ b c f i$
$ b c i i$
$ b $
```

SUCCESS

Lab 7 Output:

Lab 8 Output:

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ gcc Lab-8.c
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
```

SHIFT REDUCE PARSER

GRAMMER

$E \rightarrow E + E$

$E \rightarrow E / E$

$E \rightarrow E * E$

$E \rightarrow a/b$

enter the input symbol: a+b

stack implementation table		
stack	input symbol	action
\$	a+b\$	--
\$a	+b\$	shift a
\$E	+b\$	$E \rightarrow a$
\$E+	b\$	shift +
\$E+b	\$	shift b
\$E+E	\$	$E \rightarrow b$
\$E	\$	$E \rightarrow E * E$
\$E	\$	ACCEPT

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
```

SHIFT REDUCE PARSER

GRAMMER

$E \rightarrow E + E$

$E \rightarrow E / E$

$E \rightarrow E * E$

$E \rightarrow a/b$

enter the input symbol: a*b+a

stack implementation table		
stack	input symbol	action
\$	a*b+a\$	--
\$a	*b+a\$	shift a
\$E	*b+a\$	$E \rightarrow a$
\$E*	b+a\$	shift *
\$E*b	+a\$	shift b
\$E*b	+a\$	$E \rightarrow b$
\$E	+a\$	$E \rightarrow E * E$
\$E+	a\$	shift +
\$E+a	\$	shift a
\$E+E	\$	$E \rightarrow a$
\$E	\$	$E \rightarrow E * E$
\$E	\$	ACCEPT

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ gcc Lab-9.c
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
INTERMEDIATE CODE GENERATION
```

```
Enter the Expression :a:a+b*c+d
The intermediate code:      Expression
Z := b*c                    a:a+Z+d
Y := a+Z                    a:Y+d
X := Y+d                    a:X
a := Xxuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$
```

Lab 9 Output:

Lab 10 Output:

```
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ gcc Lab-10.c
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ cat input.txt
+ a b t1
* t1 c t2
- t2 d t3
/ t3 e t4
= t4 - x
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ ./a.out
xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ cat output.txt

MOV R0,a
ADD R0,b
MOV t1,R0
MOV R0,t1
MUL R0,c
MOV t2, R0
MOV R0,t2
SUB R0,d
MOV t3,R0
MOV R0,t3
DIV R0,e
MOV t4,R0
MOV R0,t4
MOV x,R0
MOV R0,t4
MOV x,R0xuman-stha@xumanstha-L0Q-15IRH8:~/Documents/CDCPractical$ █
```